Solutions to Quiz 10

Question 1

Give examples of each of the following:

1. A convex set $S \subset \mathbb{R}^n$ with volume greater than 2^n that contains no non-zero lattice point.
2. A symmetric set $T \subset \mathbb{R}^n$ with volume greater than 2^n that contains no non-zero lattice point.
3. A symmetric convex set $U \subset \mathbb{R}^n$ containing the origin with volume 2^n that contains no non-zero lattice point.

Solution

One must assume that $n \geq 2$! (Which was not mentioned in the original question.) This is required for S and T as given below to be correct. However, one can choose alternate examples for T with $n = 1$.

$U = \{(x_1, \ldots, x_n) : |x_i| < 1; i = 1, \ldots, n\}$ is a symmetric convex set with volume 2^n that contains no lattice point other than the origin.

$T = \{(x_1, \ldots, x_n) : 1 < |x_i| < 2; i = 1, \ldots, n\}$ is a symmetric set with volume greater than 2^n that contains no lattice point.

$S = \{(x_1, \ldots, x_n) : 0 < x_1 < 2; |x_i| < 1; i = 2, \ldots, n\}$ is a convex set with volume greater than 2^n that contains no lattice point.